

1. DDRx – Sets pin direction

PORTx - Drives output level or configures input pull-up

PINx – Reads voltage level on the pin

PD2 configuration - Input with pull-up resistor enables

2. PD3 / INT3

PD1 / INT1

Jumper Wires are required because port B pins are regular pins, requiring external wires to connect to the interrupt pins

3. EICRA – Tells what triggers the interrupt

EIMSK – Tells which interrupt pin to read

SREG I-bit / SEI – Tells whether to enable interrupts

Interrupt Vector – flash memory addresses pointing to specific interrupt service routines

Rising/falling-edge selection is controlled by: EICRA

4. UP ISR: if $\text{BlinkFreq} \leq 15$, then ____Increment____ BlinkFreq

DOWN ISR: if $\text{BlinkFreq} > 1$, then ____decrement____ BlinkFreq

If $\text{BlinkFreq} = 15$ and UP is selected: $\text{BlinkFreq} = 15$

If $\text{BlinkFreq} = 1$ and DOWN is selected: $\text{BlinkFreq} = 1$

5. $\text{BlinkFreq} = 1$: PC3 PC2 PC1 PC0 = 0001

$\text{BlinkFreq} = 5$: PC3 PC2 PC1 PC0 = 0101

$\text{BlinkFreq} = 10$: PC3 PC2 PC1 PC0 = 1010

$\text{BlinkFreq} = 15$: PC3 PC2 PC1 PC0 = 1111

6. Number the following events in the order they occur.

- _1_ A rising edge occurs on the interrupt pin.
- _2_ The corresponding ISR begins.
- _3_ BlinkFreq becomes 7.
- _4_ LEDs 0–3 display 0111.
- _5_ RETI returns control to the main program.
- _6_ The main program uses the new blink rate

<https://github.com/vdnguyen1446/EECE3624-Lab05>